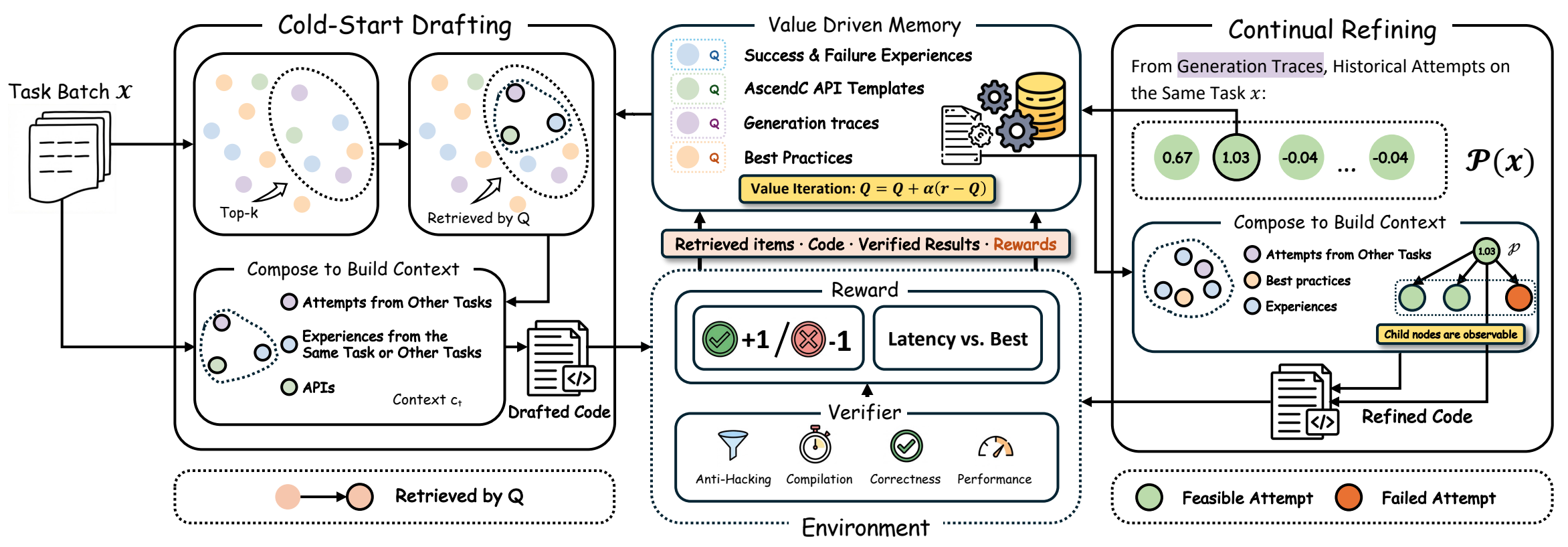
Cold-Start Drafting
Task Batch $\mathcal{X}$
Top-k
Retrieved by Q
Compose to Build Context
Attempts from Other Tasks
Experiences from the Same Task or Other Tasks
APIs
Context c_t
Drafted Code
Retrieved by Q
Value Driven Memory
Success & Failure Experiences
AscendC API Templates
Generation traces
Best Practices
Value Iteration: $Q = Q + \alpha(r - Q)$
Retrieved items · Code · Verified Results · Rewards
Reward
+1 / -1
Latency vs. Best
Verifier
Anti-Hacking
Compilation
Correctness
Performance
Environment
Continual Refining
From Generation Traces, Historical Attempts on the Same Task x:
0.67
1.03
-0.04
...
-0.04
$\mathcal{P}(x)$
Compose to Build Context
Attempts from Other Tasks
Best practices
Experiences
1.03
$\mathcal{P}$
Child nodes are observable
Refined Code
Feasible Attempt
Failed Attempt